

Association of Changes in Hip and Knee Kinematics During a Single-Leg Squat With Changes in Patient-Reported Outcomes at 6 Months and 1 Year After Hip Arthroscopy

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Background: Previous studies have demonstrated alterations in squat kinematics in patients with femoroacetabular impingement syndrome (FAIS). Little is known about the effects of arthroscopic hip surgery on biomechanics during a single-leg squat (SLS) in these patients.

Purpose/Hypothesis: The purpose of this study was to determine if (1) lower extremity dynamic range of motion (ROM) during an SLS task improves after hip arthroscopy for FAIS and (2) correlations exist between changes in patient-reported outcomes (PROs) and changes in lower extremity dynamic ROM during an SLS after hip arthroscopy for FAIS. It was hypothesized that dynamic hip ROM would improve after hip arthroscopy and that hip dynamic ROM would be associated with changes in PRO scores at both 6 months and 1 year.

Study Design: Descriptive laboratory study.

Methods: Patients with FAIS performed 3 SLSs that were analyzed using a 20-camera motion capture system. Dynamic ROMs were calculated in 3 planes for the hip, knee, ankle, and pelvic segments. Squat depth was calculated as the change in vertical center of mass during the squat cycle. PROs including the Hip Outcome Score–Activities of Daily Living (HOS–ADL), Hip Outcome Score–Sports (HOS–Sports), International Hip Outcome Tool–12, and visual analog scale for pain scores were collected preoperatively and at the time of postoperative testing. Paired-samples *t* tests were used to compare kinematic variables pre- and postoperatively. Correlations were used to compare changes in PROs with changes in kinematics. All statistical analysis was performed using SPSS Version 26.

Results: Fifteen patients were tested preoperatively and at a mean of 9 months postoperatively. All PRO measures improved postoperatively at 6 months and 1 year. Squat depth and sagittal plane hip and knee dynamic ROMs were significantly improved postoperatively. Positive correlations existed between changes in (1) hip ROM with the 6-month HOS–ADL score ($r = 0.665$) and (2) knee ROM with the 6 month ($r = 0.590$) and 1-year ($r = 0.565$) HOS–Sports scores.

Conclusion: Dynamic sagittal plane hip and knee ROMs improve after hip arthroscopy for FAIS. These improvements demonstrate strong correlations with improvements in some but not all postoperative PROs.

Clinical Relevance: The current study sought to better understand the role of dynamic movement in the diagnosis and treatment of FAIS. These findings indicate that dynamic ROM and squat depth can, similarly to PROs, serve as biomarkers for patient function both before and after hip arthroscopic surgery.

Keywords: femoroacetabular impingement syndrome; hip arthroscopy; biomechanics; motion analysis; patient-reported outcome; single-leg squat

Femoroacetabular impingement syndrome (FAIS) is a clinical hip disorder characterized by hip symptoms, clinical signs, and imaging that indicate abnormal hip morphology.^{6,7} Movement is central to both the diagnosis and treatment of FAIS. A recent international consensus statement recommended more study of movement-related factors

relevant to clinical hip disorders like FAIS.⁷ Hip biomechanics are altered in patients with FAIS when compared with healthy controls; however, it remains unknown if the biomechanical alterations found in patients with FAIS contribute to the development of, or are a result of, this clinical hip disorder.^{13,18} Hip arthroscopy has emerged as a primary treatment for FAIS. A considerable amount of clinical research has shown improvement in symptoms and patient-reported outcomes (PROs) of function after hip arthroscopy for FAIS. However, a systematic review by King and colleagues¹³ revealed inadequate evidence to draw conclusions on the effect of arthroscopic surgery on hip biomechanics. Additionally, these authors also recommended biomechanical studies of more difficult and complex tasks to help better understand movement strategies in patients with FAIS.¹³ More recent biomechanical studies have found that patients with FAIS demonstrate altered hip biomechanics and impaired single-leg squat (SLS) performance compared with controls.^{17,18} Therefore, the study of biomechanics during an SLS task in patients before and after arthroscopic surgery for FAIS will help add important information to the understanding of the role of movement in FAIS.

The SLS task is useful to evaluate dynamic movement patterns and biomechanical deficiencies in patients with FAIS.^{19,20} McGovern and colleagues¹⁹ demonstrated that the SLS is a valid and reliable test of functional performance of patients with FAIS. Additionally, in patients with FAIS, an SLS task was shown to elicit more biomechanical differences at the hip than a double-leg squat task.¹⁷ Furthermore, SLS task performance is impaired in patients with FAIS compared with controls, and the predictors of this impaired SLS task performance were hip and knee biomechanics, hip muscle strength, and femoral morphology.¹⁸ Both retrospective clinical and laboratory-based studies have used the SLS task to determine improvement in movement patterns after nonoperative treatment for hip disorders such as FAIS.^{9,10,21} However, limited evidence exists that examines hip biomechanics before and after hip arthroscopy for FAIS, and this evidence is limited to tasks such as walking and sit-to-stand activities.²³⁻²⁵

The measurement of improvement or change in function in patients with FAIS has been primarily quantified through recording PRO scores before and after hip arthroscopy.^{11,22,26,27} This early and midterm PRO clinical outcome evidence suggests that hip arthroscopy does lead to improved patient-reported function after surgery.²² However, it remains unknown if similar improvements in biomechanics occur during functional tasks in patients

undergoing hip arthroscopy for FAIS. As the field of hip arthroscopy evolves, it is important to consider changes in function from the perspective of both PROs and quantitative biomechanical joint function. Movement has always been central to the diagnosis of FAIS, with its importance becoming more apparent since the recent description of FAIS as a “motion-related clinical hip disorder.”^{5,7} Therefore, investigating biomechanics before and after surgery is important to understand how movement changes and possibly improves in patients after surgery.

Investigating the relationship between PROs and lower extremity biomechanical function before and after hip arthroscopy for FAIS will help clinicians to better understand the role that movement plays in the treatment of FAIS. The 2 aims of this study were to determine if (1) lower extremity dynamic range of motion (ROM) during an SLS task improves after hip arthroscopy for FAIS and (2) correlations exist between changes in PROs and changes in lower extremity dynamic ROM during an SLS after hip arthroscopy for FAIS. We hypothesized that dynamic hip ROM would improve after hip arthroscopy and that hip dynamic ROM would be associated with changes in PRO scores at both 6 months and 1 year.

METHODS

Study Design Overview

This study is a longitudinal laboratory-based study. Patients undergoing hip arthroscopy participated in a motion capture testing session at 2 time points, before and after surgery. Before the preoperative testing session, all participants provided informed consent for the study, which was approved by our institutional review board. The sample size was determined from a 1-tailed *a priori* power analysis to detect an effect size of 0.80, which was based on data collected from previously published studies investigating biomechanical changes before and after hip arthroscopic surgery for FAIS.^{3,16} To detect this level of within-group difference ($\alpha = .05$) at a power of at least 0.85, 13 participants were needed. To account for variability in the postoperative testing time, 15 participants who were undergoing hip arthroscopy for FAIS were recruited.

Participants

This study consisted of 15 patients >18 years of age who underwent hip arthroscopy for unilateral FAIS. These

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participants represent a subsample from a larger study that compared SLS performance and biomechanics between individuals with FAIS and healthy controls, for which the findings have been previously reported.¹⁸ The participants in the current study were patients who elected to undergo hip arthroscopy and returned for a follow-up motion capture testing session. All patients were recruited from the clinical practice of a single orthopaedic surgeon (S.J.N.) who specializes in the arthroscopic surgical treatment of hip disorders. The original study design was for follow-up testing to occur at 6 months after surgery; however, the COVID-19 pandemic affected the postoperative testing follow-up period, which occurred at a mean of 9.2 ± 2.1 months. Recruitment and testing for this study occurred from December 2018 to August 2020.

Hip Arthroscopic Surgery for FAIS

All patients underwent primary hip arthroscopy at a high-volume academic center by a fellowship-trained hip arthroscopic surgeon (S.J.N.) as previously described.^{4,8} Patients were placed in the supine position on a traction table with a well-padded perineal post. Axial traction was applied to allow access to the central compartment of the hip. Needle localization was used to establish standard anterolateral, modified midanterior, and distal anterolateral accessory portals.⁴ A transverse interportal capsulotomy was performed to establish access to the central compartment. Acetabular rim trimming and labral refixation were performed when appropriate in cases of pincer-type morphology and acetabular labral tearing using a distal anterolateral accessory and modified midanterior portal. In all cases, a vertical T-capsulotomy was made, perpendicular to the previous horizontal interportal capsulotomy and extending distally along the length of the femoral neck to the intertrochanteric line, to visualize and assess cam morphology. Arthroscopic femoral osteochondroplasty was performed, under fluoroscopic guidance, to address cam morphology. Under direct visualization and fluoroscopic guidance, dynamic examination was performed to confirm adequate and comprehensive resection of any bony impingement. The T-capsulotomy was repaired using a suture shuttling device, beginning at the base of the vertical limb, extending proximally, followed by closure of the interportal segment.¹ Postoperatively, patients were fitted with a hip brace, allowing 0° to 90° of hip flexion, and patients were limited to a crutch-assisted gait with 20 lb foot-flat weightbearing for a period of 3 weeks for procedures not involving microfracture. All patients were referred to physical therapy within the first postoperative week.

Postoperative Hip Arthroscopy Rehabilitation

All patients undergoing hip arthroscopy followed a movement control-based rehabilitation progression over 3 phases of postoperative physical therapy.¹⁵ Although phases of any rehabilitation progression are not mutually exclusive, a standard set of criteria were used to advance progressions through 3 phases (see Appendix Table A1,

TABLE 1
Characteristics of Patients Undergoing Hip Arthroscopy for FAIS and Postoperative PT Rehabilitation Treatment Information^a

Patients With FAIS (n = 15)	
Patient sex, female/male	9/6 (69.2)
No. of left involved side	7 (46.7)
Age, y	27.9 ± 7.6
Body mass index	22.7 ± 3.0
No. of preop PT sessions	7.9 ± 7.5
Duration of preop PT, wk	6.1 ± 8.1
No. of postop PT sessions	32.5 ± 12.7
Duration of postop PT, wk	22.8 ± 13.9
Time from arthroscopic hip surgery to postop testing, mo	9.5 ± 1.8

^aData are reported as n (% female) or mean $\pm$ SD. FAIS, femoroacetabular impingement syndrome; postop, postoperative; preop, preoperative; PT, physical therapy.

available in the online version of this article). Although postoperative physical therapy treatments were not formally monitored throughout the course of the rehabilitation as part of this study, a summary of patient characteristics and information related to formal postoperative physical therapy treatment is included in Table 1.

SLS Motion Capture Data Acquisition

A 20-camera motion capture system (Optitrack, Natural Point Inc.) and force plates (Bertec Corp.) acquired 3-dimensional kinematic and kinetic data at 100 and 1000 Hz, respectively, using a marker set previously described in detail.¹⁸ A static standing calibration trial was recorded before any movement trials to define body segment parameters based on the marker positions from this trial. After the static trial, all participants then performed 3 individual trials of an SLS task using a self-selected strategy previously described in detail.¹⁸

Motion Analysis Data Processing and Analysis

All raw 3-dimensional kinematic marker position and kinetic force data were processed using Visual3D (C-Motion Inc) as previously described in detail.^{17,18} The model segment positions were expressed relative to the global coordinate system, and joint angles were expressed as motion of the distal segment in relation to the proximal segment. Joint center locations for the hip, knee, and ankle were determined based on marker position as previously described.¹⁸ A Cardan rotation sequence was used to express joint flexion-extension around the mediolateral *x*-axis, abduction-adduction around the anteroposterior *y*-axis, and internal-external rotation around the vertical *z*-axis. The mean resolution for skin-based marker tracking for the current study was 0.754 ± 0.04 mm, and the joint rotation resolution was 2.6° . Pelvic motion was described as anterior and posterior tilt, contralateral (nonstance side) drop and rise, and forward and

backward rotation, corresponding to rotation around the *x-y-z* axes of the laboratory, respectively. The squat cycle was defined based on the change in the vertical position of the center of mass marker, which has been previously defined in detail elsewhere.^{17,18} The biomechanical variables of interest extracted from non-time normalized waveforms included hip, knee, ankle, and pelvic dynamic ROMs during the squat cycle. Dynamic ROM was the difference between the maximum and minimum lower extremity joint angles and pelvic segment angles. Additionally, squat depth was the difference between the center of mass marker height at the stationary standing position before the squat and the lowest vertical position during the squat. To determine the reliability of all kinematic variables, a preliminary study was conducted in our laboratory using a test-retest design. Intraclass correlation coefficients (ICC_{2,k} values) were calculated to determine the reliability of the kinematic variables of interest. An excellent correlation was defined as an ICC > 0.75. The ICC_{2,k} values for all variables were between 0.77 and 0.94, indicating excellent reliability of the kinematic measures in the current study.

Hip-Specific PROs

Patients undergoing hip arthroscopy for FAIS completed hip-specific PROs for activity and quality of life. The Hip Outcome Score–Activities of Daily Living (HOS–ADL), Hip Outcome Score–Sports, (HOS–Sports) and International Hip Outcome Tool–12 scores were collected preoperatively and at the 6-month and 1-year postoperative time points. The change scores were reported between the preoperative time point and the 6-month and 1-year time points as the delta 6-month and delta 1-year scores, respectively.

Statistical Analysis

All continuous variable distributions were assessed for normality using the Shapiro-Wilk test. Nonparametric tests were used to compare variables that were not normally distributed. The primary variables of interest during the single-leg task were squat depth in meters; dynamic ROMs of the hip, knee, and pelvis in the sagittal, frontal, and transverse planes; and dynamic ROM for the ankle in the sagittal and frontal planes. Paired-samples *t* tests were used to determine within-group differences for all variables in patients with FAIS before and after hip arthroscopy. If variables deviated from normality, the Friedman test was used to compare within-group differences. In cases of a significant within-group difference on a biomechanical variable, the correlations were assessed between the before and after surgery change in these variables and change in PRO scores at 6 months and 1 year using the Pearson product-moment correlation coefficient (*r*). The strengths of significant correlations were defined as weak, 0.1 to 0.29; moderate, 0.3 to 0.49; and strong, >0.5.² All statistical analyses were performed using SPSS software (Version 26; IBM Corp).

RESULTS

Preoperative and Postoperative PROs and Radiographic Measurements in Patients With FAIS

A significant improvement in all PROs was noted after hip arthroscopic surgery for FAIS (Table 2). Additionally, a significant reduction in the radiographic alpha angle was found within patients undergoing hip arthroscopy for FAIS, although no differences in acetabular morphology measurements of the lateral center-edge and Tönnis angles were found (Table 3).

Squat Performance

Patients with FAIS demonstrated a significant improvement in squat depth after hip arthroscopic surgery (*P* = .012) (Table 4). No differences in squat velocity during the descent (*P* = .692) or ascent (*P* = .697) phases were found within patients with FAIS before and after surgery (Table 4).

Hip, Knee, Ankle, and Pelvic Dynamic ROMs

Hip joint sagittal plane dynamic ROM significantly increased in patients after hip arthroscopy for FAIS (preoperative: 65.0 ± 11.7; postoperative: 71.8 ± 10.6; *P* = .007) (Figure 1A). Similarly, a significant increase in sagittal plane dynamic knee ROM was found in patients after surgery for FAIS (preoperative: 70.2 ± 9.3; postoperative: 74.7 ± 12.2; *P* = .029) (Figure 1A). No changes were found in sagittal plane ankle dynamic ROM after surgery (Figure 1A). Additionally, no changes were found in hip, knee, or ankle frontal plane dynamic ROMs (Figure 1B) or hip transverse plane dynamic ROM (Figure 1C). Similarly, the dynamic pelvic ROM remained unchanged in patients after hip arthroscopy for FAIS (Figure 1, A–C).

Association Between Changes in Dynamic ROM and 6-Month and 1-Year Change in PRO Scores

A strong correlation was found between sagittal plane delta dynamic hip ROM and the 6-month change in HOS–ADL score (*P* = .013; *r* = 0.665) (Figure 2A). Strong significant correlations were also found between sagittal plane delta dynamic knee ROM during an SLS and the change in HOS–Sports score at 6 months (*P* = .043; *r* = 0.590) (Figure 2B) and 1 year (*P* = .028; *r* = 0.565) (Figure 2C).

DISCUSSION

The main findings of the study were that hip and knee dynamic sagittal plane ROMs significantly increased after hip arthroscopy for FAIS, and these changes were strongly correlated with changes in PRO scores. Specifically, strong correlations were found between the change in sagittal plane dynamic hip ROM and the change in 6-month

TABLE 2
PROs for Patients Undergoing Hip Arthroscopy for FAIS at 6-Month and 1-Year Postoperative Time Points^a

	Preop	6 mo Postop	1 y Postop	P Value: Preop vs 6 mo	P Value: Preop vs 1 y
HOS-ADL	73.6.1 ± 12.0	92.5 ± 5.7	95.9 ± 4.0	<.001 ^b	<.001 ^b
HOS-Sports	47.1 ± 22.5	70.2 ± 27.9	82.1 ± 29.0	<.001 ^b	<.001 ^b
iHOT-12	41.7 ± 15.7	84.9 ± 12.8	89.2 ± 18.3	<.001 ^b	<.001 ^b
VAS-P	47.8 ± 18.8	10.7 ± 12.4	3.6 ± 5.3	<.001 ^b	<.001 ^b

^aData are reported as mean ± SD. FAIS, femoroacetabular impingement syndrome; HOS-ADL, Hip Outcome Score–Activities of Daily Living; HOS-Sports, Hip Outcome Score–Sports; iHOT-12, International Hip Outcome Tool–12; postop, postoperative; preop, preoperative; PRO, patient-reported outcome; VAS-P, visual analog scale for pain.

^bStatistically significant at $\alpha = .05$.

TABLE 3
Radiographic Measures Before and After Surgery for FAIS^a

	Preoperative	Postoperative	P Value
Dunn alpha angle, deg	65.7 ± 10.9	38.8 ± 3.5	<.001
Lateral center-edge angle, deg	30.8 ± 4.6	30.7 ± 4.3	.867
Tönnis angle, deg	7.5 ± 5.2	6.2 ± 5.9	.190

^aData are reported as mean ± SD. FAIS, femoroacetabular impingement syndrome.

HOS-ADL scores, as well as between the change in dynamic knee ROM and the change in the HOS-Sports score at both 6 months and 1 year. Although the increases in dynamic hip ROM were small, the positive relationship between these changes and changes in PRO scores after surgery represents functional improvement after hip arthroscopy for FAIS. A potential reason for the improvement in SLS task performance found in this study could be the result of greater functional ability in these patients after surgery because of a reduction in hip symptoms, such as pain, that previously precluded movement. However, more research is needed to elucidate the primary contributions to the changes in these biomechanical variables in patients undergoing hip arthroscopy for FAIS.

The findings of improved lower extremity biomechanical function and SLS task performance after hip arthroscopy for FAIS highlight the importance of considering movement task performance when determining treatment success. Overall, patients' SLS depth increased after hip arthroscopy for FAIS, driven by increased hip and knee sagittal plane dynamic ROMs. However, it should be noted that the mean differences in squat depth were small but at least equivalent to the minimal detectable change of 0.03 m, which was determined from the preliminary reliability data of our laboratory. Additionally, the significant changes found in dynamic hip and knee sagittal plane ROMs in the current study also exceeded the minimal detectable change values of 6.4° and 3.4° for these variables, respectively. Despite these small differences in depth, hip and knee sagittal plane dynamic motion may signify an important biomarker for patients' functional performance after surgery, considering that these variables were shown

TABLE 4
Single-Leg Squat Depth and Descent and Ascent Velocities Before and After Hip Arthroscopy for FAIS^a

	Preop FAIS (n = 15)	Postop FAIS (n = 15)	P Value
Squat depth, m	0.24 ± 0.04	0.27 ± 0.07	.012 ^b
Descent velocity, m/s	−0.32 ± 0.12	−0.31 ± 0.11	.692
Ascent velocity, m/s	0.44 ± 0.14	0.44 ± 0.11	.697

^aData are reported as mean ± SD. FAIS, femoroacetabular impingement syndrome; postop, postoperative; preop, preoperative.

^bStatistically significant at $\alpha = .05$.

to also be impaired compared with controls in another study.¹⁸

Although the findings from previous studies investigating biomechanical function before and after hip arthroscopy for FAIS were heterogeneous,^{23–25} the current investigation findings are consistent with more recent postoperative hip arthroscopy biomechanical literature.²⁵ A potential reason for the similarity with this literature is that patients were treated surgically with more advanced, and now standard, arthroscopic techniques, such as labral repair and capsular closure. For example, Samaan and colleagues²⁵ found that 7 months after hip arthroscopy for FAIS, patients demonstrated improved hip kinematic patterns during gait within the surgical limb, which is consistent with our findings of increased dynamic ROM during an SLS task. Additionally, the period between preoperative and postoperative testing was also similar between our study and that of Samaan and colleagues. Although it is important to quantify functional improvement after surgery, it is also important that future studies investigate factors that may contribute to or predict improved biomechanical function after hip arthroscopy for FAIS.

A number of factors can potentially contribute to altered biomechanics in patients diagnosed with FAIS. A recent study by Lewis and colleagues¹⁴ using dual fluoroscopy imaging to quantify hip kinematics during walking found altered intra-articular joint mechanics in patients with cam-type FAIS compared with a control group. Specifically, these authors showed that patients with cam-type FAIS exhibited larger changes in bone-to-bone distance in all

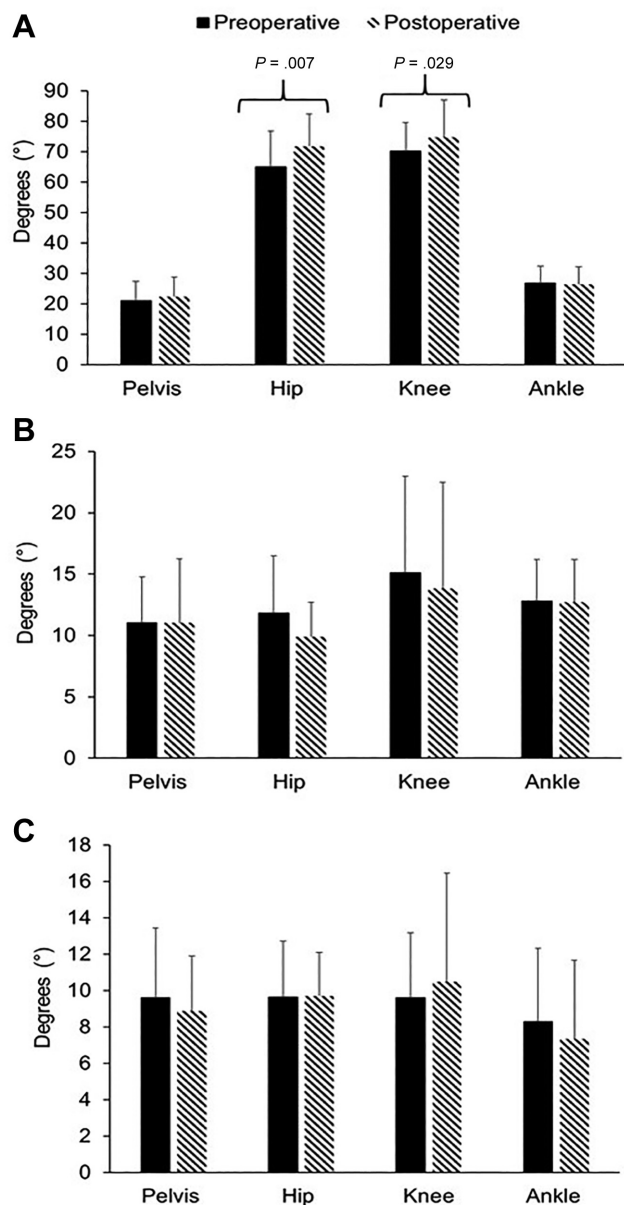


Figure 1. Pelvis and lower extremity dynamic ranges of motion in the (A) sagittal, (B) frontal, and (C) transverse planes of motion during a single-leg squat task at the preoperative (solid black) and postoperative (black dashed) time points. One-tailed statistical significance was set at an alpha level of $<.05$.

regions of the acetabulum except the anterior superior region, which is the commonly observed location of cartilage and labral injury in patients with cam morphology. Although these findings challenge the premise that altered hip mechanics are due to direct impingement during tasks that demand end-range position (ie, squatting), the translation (ie, sliding) of the femoral head within the acetabulum during dynamic activity is limited and not well understood. Additionally, each functional task performed

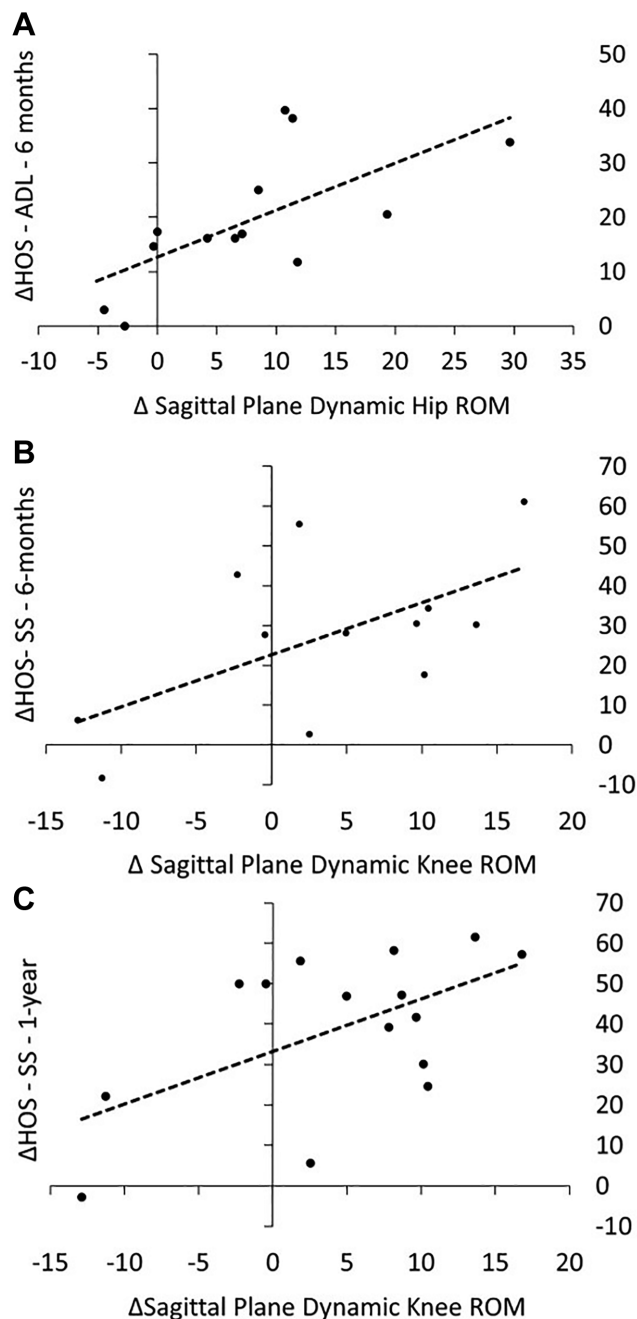


Figure 2. Bivariate correlations between the (A) change (Δ) in sagittal plane hip dynamic range of motion (ROM) and 6-month change in Hip Outcome Scale–Activities of Daily Living (HOS-ADL) score, (B) change in sagittal plane knee dynamic ROM and 6-month change in Hip Outcome Scale–Sports (HOS-SS) score, and (C) change in sagittal plane knee dynamic ROM and 1-year change in HOS–Sports score.

by patients requires different internal forces from muscle activations and soft tissue deformation related to the task demand. Therefore, intra-articular bone-to-bone kinematics are likely task-specific, and direct mechanical

impingement during certain activities that require near-end ROM is still plausible. In particular, another recent biomechanical study using motion capture techniques in patients with FAIS found that femoral alpha angles were one of several predictors of impaired SLS performance, in addition to other variables, such as reduced dynamic knee, hip, and pelvic ROMs, and hip muscle strength.¹⁸ When the current biomechanical findings are considered within the context of these previous studies,^{14,25} it seems clear that the improvements observed in biomechanics after surgery for FAIS are likely multifactorial and could be related to factors such as elimination of pain, symptomatic bony impingement, improved muscle strength, optimized motor control, and greater overall functional ability.

Patients undergoing hip arthroscopy for FAIS in the current study demonstrated significantly reduced general pain and improved function quantified using a visual analog scale pain scale and hip-specific PROs, respectively. Additionally, strong correlations were also found between the changes in hip and knee dynamic ROMs and changes in 6-month and 1-year PRO scores, supporting an overall improvement in functional ability in this sample after hip arthroscopy for FAIS. A recent study by Kierkegaard and colleagues¹² found that peak hip motion was associated with both the sports function and the quality of life domains on the Copenhagen Hip and Groin Outcome Score preoperatively and 1 year after surgery. These authors suggest that this association may signify the clinical importance of improved hip biomechanics after surgery. Perhaps the clinical importance of the relationship between PROs and biomechanical function in patients undergoing hip arthroscopy for FAIS is that it represents a neuromuscular readjustment period during a patient's recovery. For example, as factors such as hip motion, muscle function, motor control, and functional performance increase after removal of the painful impingement stimuli and repair of damaged tissues after hip arthroscopy, the system likely requires significant time to adjust to these dynamic changes. Therefore, as patients progress through postoperative hip arthroscopy rehabilitation it is important that improvement occurs in both joint-level biomechanical function and patient-reported function to obtain a successful postoperative outcome.

Overall, patients demonstrated a significant reduction in pain at both the 6-month and the 1-year postoperative time points (see Table 2). A visual analog pain scale measured the patient's mean pain over the past week, which provides a limited assessment of the complex pain experience and does not assess the relationship between pain and task performance. In the current study, 1 patient did demonstrate worse SLS task performance and a reduction in dynamic ROM for all variables. Additionally, this patient had greater pain and lower PRO scores at 6 months as well as a <10-mm improvement on the visual analog scale at 1 year. Although this is a single patient example from this sample, it highlights the complexity of the relationship among hip pain, function, and movement in patients with FAIS. Future studies should investigate task-specific pain in patients with FAIS to determine the role that hip pain may play in movement alterations or improvement in patients receiving treatment for FAIS.

Limitations

The findings of this study must be considered within the context of a number of limitations. The original study design was for patients to return at 6 months for follow-up testing as this time frame would account for a completed course of postoperative rehabilitation. However, the period for follow-up testing was between 6 months and 1 year after surgery, which introduced more variability into the study sample as the potential activity levels between the extremes of these postoperative time points could be quite different. This limitation was due to the suspension of postoperative testing because of the COVID-19 pandemic. To account for the added sample variability of postoperative testing time, 2 additional participants were recruited beyond the calculated study sample size from the a priori power analysis. Despite the adequate sample size for the biomechanical comparisons, the study was not powered to control for the potential influence of biological sex, which has been shown to influence biomechanics. Future prospective studies with larger samples sizes are planned to investigate the influence of other factors on biomechanical changes such as sex, age, tissue damage, and bony morphology. The postoperative physical therapy rehabilitation for the sample was not well controlled on an individual basis, as multiple therapy providers were used, which could ultimately influence the patient's postoperative ability. However, all patients completed 3 months of structured physical therapy care, per a standardized protocol, and were provided a detailed postoperative rehabilitation guideline that is summarized in Appendix Table A1 (available online) to help inform treatment progression. Finally, the inherent limitation of motion artifact from skin-based mounted markers should be considered, which can be exacerbated during fast or rapid movements. However, the data from our laboratory demonstrate very good to excellent reliability for all variables, and excellent reliability for all variables in the study that demonstrated significant differences.

CONCLUSION

Hip and knee sagittal plane dynamic ROMs during an SLS task increased after hip arthroscopy for FAIS, which led to a small improvement in SLS depth. Additionally, the amount of change in hip and knee dynamic ROMs was strongly associated with the amount of change in some PRO scores at 6 months and 1 year after hip arthroscopy for FAIS.

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